Chapter 11

Strategic Cost Management

Learning Objectives

- Understand the importance of total cost of ownership (TCO) in procurement decisions
- Distinguish between price analysis and cost analysis
- Apply cost reduction strategies including value analysis and target costing
- Understand learning curves and their impact on cost projections
- Calculate and interpret break-even analysis and rate of return

CASE STUDY: The $50 Million Mistake

How a global electronics manufacturer lost millions by focusing only on unit price

Based on real procurement scenarios in the electronics industry

Case Background: GlobalTech Inc.

The Situation: GlobalTech sources 500,000 custom PCB assemblies annually

- Current supplier (Supplier A): $48/unit, 99.2% quality, 2-week lead time
- New supplier (Supplier B): $42/unit, promises same quality and delivery
- Potential savings: $3 million per year

The Decision:

- Procurement team recommended switching to Supplier B
- Based purely on unit price comparison (price analysis)
- Management approved the switch

What Went Wrong? The Hidden Costs

- Quality dropped to 96.8% → $1.2M in defect costs
- Lead time stretched to 4 weeks → $800K in expediting fees
- 3 supply disruptions in Year 1 → $600K in line shutdowns
- Engineering support was poor → $400K in rework
- Switching costs (tooling, qualification): $500K one-time

Total HIDDEN costs: $3.5M/year + $500K switching
Net result: $50M+ loss over 10-year product lifecycle
The '$6/unit savings' actually COST $7/unit more

Lesson: Toyota's Target Costing Approach

Real-World Example: Cost Management Success

Toyota's Philosophy:

"Cost reduction must be the responsibility of all employees, not just the purchasing department"

Target Costing Process:

- Market research determines target selling price
- Subtract desired profit margin to get target cost
- Work backwards with suppliers to achieve target cost through:
    - *Design optimization*
    - *Process improvements*
    - *Supplier collaboration and innovation*

Result: 20-30% cost reduction while improving quality

Total Cost of Ownership (TCO)

TCO is the total cost of acquiring, using, and disposing of a product or service over its entire lifecycle

Why TCO matters:

Purchase price is often only 20-40% of total lifecycle cost

Hidden costs can make a "cheap" supplier very expensive

Better decisions when all costs are visible

TCO Components:

Pre-transaction costs: Research, qualification, tooling

Transaction costs: Purchase price, delivery, customs

Post-transaction costs: Quality issues, maintenance, disposal

TCO Calculation Framework

TCO = Purchase Price + Acquisition Costs + Usage Costs + Maintenance Costs + Disposal Costs

Purchase Price	Base unit cost × quantity
Acquisition	Supplier search, qualification, tooling, training
Usage	Operating costs, utilities, consumables
Maintenance	Repairs, spare parts, downtime costs
Disposal	Removal, recycling, environmental compliance

TCO Example: Comparing Two Suppliers

Cost Category	Supplier A	Supplier B
Unit Price (10,000 units)	$100	$85
Quality Cost (defect rate)	$5/unit	$12/unit
Delivery Cost	$3/unit	$8/unit
Inventory Carrying Cost	$2/unit	$6/unit
Total Cost per Unit	**$110**	**$111**
Total Cost (10,000 units)	**$1,100,000**	**$1,110,000**

Conclusion: Despite 15% lower unit price, Supplier B costs MORE due to quality and logistics issues

Price Analysis vs. Cost Analysis

Price Analysis

When to Use:

Standard products available from multiple suppliers

Competitive bidding situations

Historical price data available

Methods:

Compare supplier quotes

Historical price trends

Market price indices

Published price lists

Advantage: Quick and simple

Limitation: Doesn't reveal cost drivers

Cost Analysis

When to Use:

Custom/engineered products

New products with no price history

Sole source situations

Need to understand cost drivers

Methods:

Break down product into cost elements

Analyze materials, labor, overhead

Should-cost modeling

Open-book negotiations

Advantage: Reveals cost drivers

Limitation: Time and expertise required

Cost Breakdown Structure

Cost Category	Description
Direct Materials	Raw materials, components, purchased parts
Direct Labor	Touch labor directly involved in production
Manufacturing Overhead	Indirect labor, utilities, depreciation, maintenance
General & Administrative	Management, sales, finance, HR overhead
Profit Margin	Supplier's desired profit percentage

Should-Cost Modeling

A detailed cost estimate built from the ground up to determine what a product SHOULD cost to manufacture

Step 1: Break down the product
Identify all components, materials, and processes

Step 2: Estimate material costs
Use market prices for raw materials and components

Step 3: Estimate labor costs
Calculate touch labor hours × standard labor rates

Step 4: Estimate overhead
Apply appropriate overhead rates (typically 50-200% of labor)

Step 5: Add profit margin
Industry-standard profit margins (typically 10-20%)

Step 6: Compare to supplier quote
Gap analysis reveals negotiation opportunities

Value Analysis / Value Engineering

Systematic method to improve the value of products by examining function and eliminating unnecessary costs

Value = Function / Cost

Key Questions to Ask:

What does this part/feature DO? (Function)

Is this function necessary?

Can the function be achieved differently at lower cost?

Can we standardize or simplify?

Can we use alternative materials?

Example: Automotive manufacturer saved $2M annually by:

Replacing custom fasteners with standard parts

Eliminating redundant quality inspections

Redesigning bracket to use 30% less material

Target Costing

Cost management approach where desired profit is subtracted from market price to determine target cost

1. Market Research	Determine competitive selling price: $100
2. Subtract Profit	Desired profit margin (20%): -$20
3. Target Cost	Maximum allowable cost: $80
4. Cost Gap Analysis	Current cost: $95 → Gap: $15 must be eliminated
5. Cross-Functional Team	Design, engineering, procurement work together
6. Achieve Target	Through design changes, supplier negotiations, process improvements

Learning Curves

As cumulative production volume doubles, the labor hours required per unit decrease by a constant percentage

Key Concepts:

80% learning curve: Each doubling of volume reduces time by 20%

90% learning curve: Each doubling reduces time by 10%

Lower % = steeper learning (more improvement)

Applies to: Complex assembly, new processes, labor-intensive work

Less applicable to: Automated processes, mature products

Example: 80% Learning Curve

Unit	Hours
1	100
2	80
4	64
8	51.2
16	41.0

Learning Curve Formula

$$Y_n = Y_1 \times n^b$$

where b = log(learning rate) / log(2)

Y_n = Time required for the nth unit

Y_1 = Time required for the first unit

n = Unit number

b = Learning curve exponent

Example: 80% learning curve, first unit takes 100 hours

$b = \log(0.80) / \log(2) = -0.322$

Time for 10th unit: $Y_{10} = 100 \times 10^{(-0.322)} = 47.7$ hours

Time for 100th unit: $Y_{100} = 100 \times 100^{(-0.322)} = 22.8$ hours

Learning Curve Applications in Procurement

1. Price Negotiations

Challenge supplier pricing based on expected learning

Negotiate price reductions for follow-on orders

Build learning curve assumptions into long-term contracts

2. Make-or-Buy Decisions

Consider learning effects when comparing internal vs external production

New product startup may have high initial costs but improve quickly

3. Production Planning

Forecast labor requirements more accurately

Plan for workforce training and skill development

4. Cost Estimation

Estimate costs for new product launches

Project total program costs over product lifecycle

Understanding Market Structures

Perfect Competition

- Many buyers and sellers
- Identical products
- Easy entry/exit
- Price takers
- Example: Agricultural commodities

Monopolistic Competition

- Many sellers
- Differentiated products
- Some price control
- Easy entry/exit
- Example: Restaurants, retail

Oligopoly

- Few large sellers
- Products may be similar or different
- Significant barriers to entry
- Price leaders
- Example: Automotive, airlines

Monopoly

- Single seller
- Unique product
- High barriers to entry
- Price maker
- Example: Utilities (regulated)

Market Structure Implications for Procurement

Perfect Competition:

Focus on price analysis (market prices readily available)

Multiple sourcing easy

Switching costs low

Monopolistic Competition:

Evaluate value of product differentiation

Consider TCO, not just price

Build relationships with preferred suppliers

Oligopoly:

Strategic supplier relationships critical

Cost analysis more important than price analysis

Long-term contracts and partnerships

Potential for supplier power - need counterstrategies

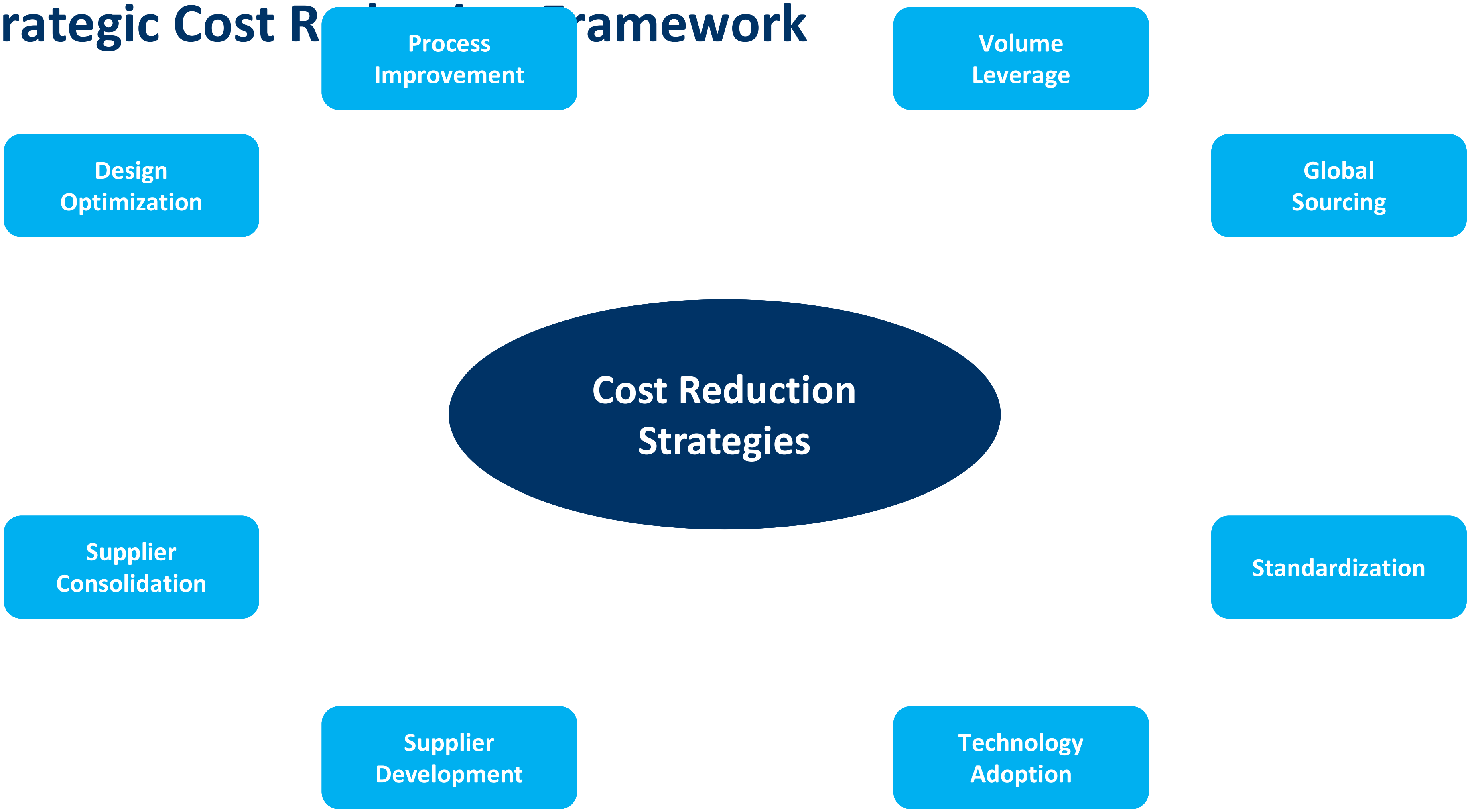
Monopoly:

Limited negotiation leverage

Focus on contract terms, service levels

Consider backward integration if feasible

Strategic Cost R[illegible]ramework



Design for Cost (DFC)

Design decisions determine 70-80% of product costs

Part Count Reduction

Fewer parts = lower assembly, inventory, and quality costs

Design for Manufacturing

Simplify production processes, reduce tooling costs

Design for Assembly

Easy assembly reduces labor time and errors

Material Selection

Balance performance requirements with material costs

Standardization

Use common components across product lines

Modular Design

Reusable modules reduce development and production costs

Supplier Consolidation Strategy

Benefits:

Volume leverage for better pricing

Reduced administrative costs

Stronger supplier relationships

Simplified quality management

Lower inventory carrying costs

Reduced transaction costs

Risks:

Increased supply risk (fewer sources)

Supplier complacency

Loss of competitive tension

Potential capacity constraints

Higher switching costs

Best Practice: Maintain 2-3 suppliers for critical items to balance leverage and risk

Global Sourcing Considerations

Cost Advantages

- Lower labor costs
- Access to specialized capabilities
- Access to raw materials
- Economies of scale
- Currency advantages

Hidden Costs

- Transportation and logistics
- Inventory carrying costs (longer lead times)
- Quality issues and rework
- Communication challenges
- Currency exchange risk
- Import duties and compliance
- Intellectual property risks

Always calculate TCO - the lowest unit price may not be the lowest total cost

Process Improvement for Cost Reduction

Lean Manufacturing

Eliminate waste in all forms (overproduction, waiting, transportation, etc.)

Just-in-time production reduces inventory costs

Continuous improvement (Kaizen)

Six Sigma

Reduce variation and defects

DMAIC: Define, Measure, Analyze, Improve, Control

Lower quality costs

Automation

Reduce labor costs

Improve consistency and quality

Increase throughput

ROI analysis critical: high upfront investment

Break-Even Analysis

Break-even point: The volume at which total revenue equals total cost (no profit, no loss)

Break-Even Quantity = Fixed Costs / (Price - Variable Cost per Unit)

Example: Outsourcing Decision

Option A (Make): Fixed costs = $100,000/year, Variable cost = $15/unit
Option B (Buy): Fixed costs = $0, Purchase price = $25/unit

Break-even: $100,000 / ($25 - $15) = 10,000 units

Decision: If volume < 10,000 units → Buy
If volume > 10,000 units → Make

Rate of Return on Investment (ROI)

Measures the profitability of an investment relative to its cost

ROI = (Net Benefit / Investment Cost) × 100%

Example 1: Quality Improvement

Investment: $50,000 in new inspection equipment
Annual savings: $25,000 (reduced defects)
ROI = ($25,000 / $50,000) × 100% = 50%
Payback period = 2 years

Example 2: Supplier Development

Investment: $100,000 in supplier training
Annual cost reduction: $40,000
ROI = ($40,000 / $100,000) × 100% = 40%
Payback period = 2.5 years

Net Present Value (NPV) and Payback Period

Net Present Value (NPV)

Accounts for time value of money: $1 today is worth more than $1 tomorrow

Formula: NPV = Σ [Cash Flow / (1 + r)^t] - Initial Investment

where r = discount rate, t = time period

Decision rule: Accept project if NPV > 0

Payback Period

Time required to recover initial investment

Formula: Payback Period = Initial Investment / Annual Cash Flow

Simpler than NPV but ignores time value of money

Useful for quick assessment of project risk

Metric	Strengths	Weaknesses
NPV	Considers time value of money	Requires discount rate estimate

Capital Budgeting for Procurement

Process of evaluating and selecting long-term investments that align with organizational goals

Procurement Capital Investments:

Make vs. buy decisions (building manufacturing capability)

Supply chain technology investments (ERP, e-procurement systems)

Supplier development programs

Vertical integration decisions

Distribution center investments

Quality management systems

Automation and robotics

Key Considerations:

Strategic alignment with organizational objectives

Risk assessment (market, technology, operational risks)

Cash flow projections over project lifetime

Comprehensive Cost Analysis Example

Scenario: Evaluating automation equipment purchase

Equipment cost: $500,000 | Installation: $50,000 | Annual maintenance: $20,000

Current labor cost: $200,000/year | Equipment reduces labor by 80%: New labor cost $40,000/year

Equipment life: 10 years | Discount rate: 10%

Financial Analysis:

Initial Investment: $500,000 + $50,000 = $550,000

Annual savings: $200,000 - $40,000 - $20,000 = $140,000

Simple Payback Period: $550,000 / $140,000 = 3.93 years

ROI (Year 1): ($140,000 / $550,000) × 100% = 25.5%

NPV (10 years, 10% discount): $309,450 (Positive - Accept project!)

Internal Rate of Return (IRR): 22.3%

Interpretation: Strong investment - positive NPV, acceptable payback period, ROI exceeds cost of capital

Cost Management Best Practices

1. Use TCO, not just purchase price
2. Engage suppliers early in design
3. Conduct should-cost analysis
4. Build learning curves into contracts
5. Standardize where possible
6. Invest in supplier development
7. Use data and benchmarking
8. Cross-functional collaboration

Remember: Every dollar saved in procurement flows directly to the bottom line

Cost Management Implementation Roadmap

1. Assess Current State	Analyze current spending, supplier base, processes
2. Identify Opportunities	TCO analysis, should-cost models, benchmarking
3. Prioritize Initiatives	High-impact, low-risk projects first
4. Develop Action Plans	Assign ownership, timelines, resources
5. Execute & Monitor	Track KPIs, measure results, adjust as needed
6. Sustain & Improve	Continuous improvement culture, share best practices

Key Cost Management Metrics

Cost Savings
Year-over-year reduction in spend (hard savings)

Cost Avoidance
Prevented cost increases (soft savings)

Procurement ROI
Value delivered / procurement department costs

Supplier Performance
Quality, delivery, cost improvements

TCO Reduction
Total cost reduction across product lifecycle

Maverick Spend
% of spend outside contracted suppliers (target: <5%)

Contract Compliance
% of spend under contract (target: >90%)

Common Cost Management Pitfalls

Focusing only on price, ignoring TCO

Short-term savings at expense of long-term relationships

Over-consolidating suppliers (excessive supply risk)

Ignoring supplier financial health (bankruptcy risk)

Inadequate change management (resistance from stakeholders)

Cutting costs that impact quality or innovation

Failing to measure and track results

Not involving cross-functional teams (engineering, quality, operations)

Remember: Not all cost reductions are created equal. Balance short-term savings with long-term value.

Technology Enablers for Cost Management

E-Procurement Systems

Automate requisition-to-pay process

Enforce contract compliance

Reduce maverick spend

Spend Analytics

Visibility into spending patterns

Identify consolidation opportunities

Track savings and performance

Supplier Portals

Real-time collaboration with suppliers

Performance tracking and scorecards

Reduce communication costs

AI and Machine Learning

Predictive analytics for cost forecasting

Automated should-cost modeling

Intelligent supplier matching

Chapter 11 Summary

Strategic cost management goes beyond price negotiations

Total Cost of Ownership provides the complete picture

Cost analysis reveals cost drivers and negotiation opportunities

Value analysis and target costing drive systematic cost reduction

Learning curves help predict and manage costs for new products

Financial analysis tools (break-even, ROI, NPV) support investment decisions

Implementation requires cross-functional collaboration and executive support

Technology enables better data, analytics, and decision-making

Balance cost reduction with quality, innovation, and supplier relationships

Discussion: How can your organization improve its cost management practices?

References & Further Reading

Recommended Resources:

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Thank You

Questions?